

Kishor Timsina, PhD, Structural Engineering (The University of Tokyo)

Asst. Professor, Madan Bhandari University of Science and Technology (MBUST)

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QUALIFICATIONS

Bachelor's Degree in Civil Engineering January 2009 – December 2012

Institute of Engineering, Pulchowk Campus, Tribhuvan University, <http://www.ioe.edu.np/> Final Year Thesis on “Earthquake Resistant Building Analysis and Design of Multistorey Building”.

Supervisor: Dr. Rajan Suwal, Institute of Engineering, Pulchowk Campus.

Master's Degree in Civil Engineering (September 2017 – September 2019) The University of Tokyo, Japan, <https://www.u-tokyo.ac.jp/en/>

Masters' Thesis on “An Optimized Retrofitting of Soft Storey RC Buildings in Nepal - A SocioTechnical Approach using Numerical Optimization”

Supervisor: Professor Kimiro Meguro, The University of Tokyo.

Doctoral Degree in Civil Engineering (September 2020 – September 2023) The University of Tokyo, Japan, <https://www.u-tokyo.ac.jp/en/>

Doctoral Thesis on “Development of Model Updating in 3D AEM to Improve the Numerical Modeling of Existing RC Frame Buildings”

Supervisor: Professor Kimiro Meguro, The University of Tokyo.

PUBLICATIONS

Journals

- **Timsina Kishor**, Chaitanya Gadagamma Krishna, and Kimiro Meguro. "Sociotechnical evaluation of the soft story problem in reinforced concrete frame buildings in Nepal." Journal of Performance of Constructed Facilities 35, no. 4 (2021): 04021019.
- **Timsina Kishor**, Chaitanya K. GADAGAMMA, Muneyoshi NUMADA, and Kimiro MEGURO. "Development of a numerical optimization framework for solving soft-storey problem in reinforced concrete frame buildings." Seisan Kenkyu 71, no. 4 (2019): 813-823.
- **Timsina Kishor**, Chaitanya Krishna GADAGAMMA, and Kimiro MEGURO. "A comprehensive field study to examine the complications in soft storey problem in Nepal." Seisan Kenkyu 71, no. 6 (2019): 1059-1063.
- Hossain, Md Shakhawat, Muneyoshi Numada, Momtahina Mitu, **Kishor Timsina**, Chaitaniya Krisna, Md Zillur Rahman, ASM Maksud Kamal, and Kimiro Meguro. "Simplified engineering geomorphic unit-based seismic site characterization of the detailed area plan of Dhaka city, Bangladesh." Scientific Reports 13, no. 1 (2023): 11151.
- **Timsina Kishor**, Nikshan Amatya, Chaitanya K. Gadagamma, and Kimiro Meguro. "Retrofitting solution for soft story mitigation in reinforced concrete frame buildings: A socio-technical approach using numerical optimization." Journal of Earthquake and Tsunami 18, no. 02 (2024): 2350040.
- Multazam, Zamzam, Kenjiro Yamamoto, **Kishor Timsina**, Chaitanya Krishna Gadagamma, and Kimiro Meguro. "Shaking table tests of a one-quarter scale model of concrete hollow block masonry houses retrofitted with fiber-reinforced paint." Scientific reports 14, no. 1 (2024): 8041.
- Htin, Lin, **Kishor Timsina**, Chaitanya Krishna Gadagamma, Qudeer Hussain, Ali Ejaz, Panumas Saingam, Wasim Khaliq et al. "Enhancing Seismic Resilience of Soft-Storey Buildings through Optimized Reinforced Concrete Infill Solutions." Engineered Science (2025).

- Multazam, Zamzam, Kenjiro Yamamoto, **Kishor Timsina**, Rajasekharan Shanthanu, and Kimiro Meguro. "Enhancing seismic resilience in weak masonry units: the impact of rebar reinforcement in concrete hollow block masonry structures." *Journal of Disaster Science and Management* 1, no. 1 (2025): 4.
- Hossain, Md Shakhawat, **Kishor Timsina**, Chaitanya Krishna Gadagamma, Muneyoshi Numada, ASM Maksud Kamal, and Kimiro Meguro. "Upgradation of 2D applied element method (AEM) for soil-structure interaction (SSI)." *Progress in Engineering Science* 2, no. 2 (2025): 100071.
- **Kishor Timsina**, Chonlasit Phumsrang, Chaitanya Krishna Gadagamma, Kimiro Meguro, "Sensitivity Based Model Updating of Operational Modal Data in 3D Applied Element Method (3DAEM)", *Engineered Science*, 2025 (Under Revision).
- **Kishor Timsina**, C. Krishna, K. Meguro. "Damage Detection of an Experimental Model Using Sensitivity-Based Model Updating with Operational Modal Data" (Drafted).
- Sneha Shrestha, **Kishor Timsina**, C. Krishna. "Development of A Mechanistic Model to Evaluate Soft Storey Phenomenon in Reinforced Concrete (RC) 2D Frame with Infill Masonry Walls" (Drafted).

Conferences and Talks

- **Kishor Timsina** "Integrating Machine Learning for Model Updating in Applied Element Method and Vibration Based Structural Health Monitoring", NEA Talk Program, Nepal Engineers Association, Nepal, August, 2025.
- **Kishor Timsina**, Chaitanya Krishna Gadagamma "Condition Assessment of RC Buildings Using: 3D AEM, Model Updating and Artificial Neural Network (ANN)", International Summer Program 2025 Nanjing Technological University, July, 2025.
- **Kishor Timsina**. "Build with Bamboo", Panellist of 1st National Bamboo Conference 2081 PreEvent at Kathmandu, 6th February, 2025.
- Shrestha, S., **Timsina, K.** and Gadagamma, C. "Development of A Mechanistic Model to Evaluate Soft Storey Phenomenon in Reinforced Concrete (RC) 2D Frame with Infill Masonry Walls", The 1st International Symposium on One Health, One World, December 2022.
- M.S. Hossain, M. Mitu, **K. Timsina**, C. Krishna, M. Numada, K. Meguro, "Simplified Engineering Geomorphic Unit Based Site Characterization of Dhaka City DAP Area", The 1st International Symposium on One Health, One World, December 2022.
- **Timsina Kishor**, Chaitanya Gadagamma Krishna, and Kimiro Meguro, "Sensitivity Based Structural Model Updating of Idealized Shear Frames using Operational Modal Data", The 1st International Symposium on One Health, One World, December 2022.
- **Timsina Kishor**, Chaitanya Gadagamma Krishna, Kenjiro Yamamoto and Kimiro Meguro, "A Comparative Study of Sensitivity and Bayesian Method for Model Updating of a 5-Degree of Freedom System", The 41st Annual Conference of the Japan Society for Natural Disaster Science, September 2022.
- **Timsina Kishor**, Chaitanya Gadagamma Krishna, Hideyomi Gokon and Kimiro Meguro, "A Proposal for An Optimized Solution for Open Storey Problem in Non-engineered Infilled Masonry Reinforced Concrete Framed Buildings in Nepal", 17th symposium on new technology for Urban Safety for Mega Cities in Asia, USMCA, December 2018.
- **Timsina Kishor**, Chaitanya Gadagamma Krishna, Hideyomi Gokon and Kimiro Meguro, "An Optimized Solution for Soft Storey Problem of Non-engineered RC Frame Buildings in Nepal", International Workshop on Disaster Mitigation and Management, IWDMM 2018, 10-11 December 2018, Visakhapatnam, India

- **K. Timsina**, S. Pradhan, S. Shrestha, “Building Code Implementation- Way to Safer Society” in 14th symposium on new technology for Urban Safety for Mega Cities in Asia, USMCA 2015.

SCHOLARSHIPS AND AWARDS

- 2009 Jan.-2012 Dec.: **Full Scholarship (Regular) in Bachelor’s in Civil Engineering, TU**
- 2009 Jan.-2012 Dec.: **The college fellowship Scholarship and Freeship in Bachelor’s in Civil Engineering, TU**
- 2017 Sep.-2019 Sep.: **ADB-JSP Scholarship for Masters’ Degree, The University of Tokyo**
- 2019 Sep.: **Furuichi-Kimitake Prize for the Outstanding Master’s Thesis, The University of Tokyo**
- 2020 Sep.-2023 Sep.: **MEXT Scholarship for Doctoral Degree, The University of Tokyo**

SKILLS

- Analysis and Design of Structures using Finite Element Modeling Tools
- Seismic Vulnerability Assessment
- Rapid and Detailed Post Earthquake Damage Assessment
- Rapid and Detailed Post Fire Damage Assessment
- Retrofitting Analysis, and Design of Structures
- Development of 2D and 3D Applied Element Method Tool in GFORTRAN
- Analysis and Design of Structures using 2D and 3D Applied Element Method
- Vibration Based Model Updating Techniques
- Structural Health Monitoring
- Programming Languages: FORTRAN, MATLAB, Python
- Use of Artificial Neural Network in structural model updating
- Numerical optimization for retrofitting of buildings
- Implementation of Civil Engineering and Disaster Risk Reduction projects
- Coordination and collaboration with different stakeholders
- Conduction of trainings, workshops and meetings

EXPERIENCES

March 2013-September 2017

Civil Engineer/ Program Coordinator

National Society for Earthquake Technology-Nepal (NSET), <https://www.nset.org.np/>

Responsibilities:

- Eastern team co-ordinator of Building code Implementation Program in Municipalities of Nepal.
- Work as one of a team leader of detail damage assessment in earthquake affected areas of Nepal after a massive 7.6M Earthquake.
- Perform analysis, design, and construction of earthquake resistant buildings.
- Perform seismic vulnerability assessment followed by analysis and design for retrofitting of existing buildings.
- Conduct orientation programs for communities on earthquake basics, preparedness, and earthquake resistant building construction.
- Conduct trainings for engineers on basic concept of earthquake resistant building design and construction, building code compliance, and seismic vulnerability assessment.
- Conduct trainings for masons on retrofitting techniques and earthquake resistant building construction.
- Assist municipalities on developing building permit process and assist on checking building

designs and drawings for building code compliance.

- Trainer for Training for Instructor “TFI” and Training of Trainers (ToT) of “Mason Training on Earthquake Resistant Building Construction”.
- Trainer for numerous Emergency Response Trainings like Community Light Search and Rescue-1001, Detail Damage Assessment Training, Rapid Visual Assessment of Buildings etc.

November 2019-November 2020

Structural Engineer/ Deputy Program Manager

National Society for Earthquake Technology-Nepal (NSET), <https://www.nset.org.np/>

Responsibilities:

- Co-ordinate and supervise the team working for “Technical Support for Resilient Community” Program for MultiHazard Risk Reduction in Rural Municipalities of Nepal.
- Co-ordination with local government, and non-governmental organizations during implementation of program.
- Analysis and design of earthquake resistant buildings and retrofitting of buildings.
- Support rural municipalities for multi-hazard risk assessment and develop disaster risk reduction and management plans and policies.
- Support in developing the framework of Building Code implementation initially for Rural Municipalities that can be later replicated to other rural municipalities in the entire country. Further, assist rural municipalities in developing building and land use regulations.
- Develop organized and systematic Building Permit System and building code implementation mechanism appropriate for rural context of Nepal. The program supports at least 7 Rural Municipalities to establish Building Permit System (BPS) in these municipalities.
- Conducting training for Engineers on Earthquake Resistant Building Design, Construction and Building Code compliance checking.
- Support in implementation of risk reduction and preparedness activities.
- Support on implementing Risk Sensitive Land Use Plan (RSLUP) in the rural municipalities.
- Provide technical assistance for demonstration of retrofitting and risk mitigation work.
- Assist in establishing Community Disaster Preparedness Learning Centers (CDPLC) in the rural municipalities.
- Technical assistance for different awareness and capacity enhancement programs.

March 2024- Onwards

Assistant Professor and Coordinator, Sustainable and Resilient Infrastructure Program, Madan Bhandari University of Science and Technology (MBUST),

Member of Executive Council, Madan Bhandari University of Science and Technology (MBUST);

<https://www.mburst.edu.np/>

RESEARCH AREA:

In the field of Structural/ Earthquake Engineering.

- Performance of Existing Structures (Masonry, RCC, Forest and Bio-Materials (bamboo, timber), Rural Structures, and Heritage Structures) (Experimental and Numerical Study)
- Structural Health Monitoring and Numerical Model Updating
- Retrofitting Methodologies for Existing Vulnerable Structures (Experimental and Numerical Study)
- Development, Improvement and Implement of Applied Element Method (AEM) for different structural engineering problems.
- Soil Structure Interaction (SSI)
- Finite Element Analysis of Structures

- Multi-Hazard Risk Reduction

Research Supervisions:

Ongoing Research Supervisions (At MBUST):

1. Supervisor of PhD student (Mukesh Maharjan) for “Characterization of Bambusa Balcooa (Dhanu Bans) and Bambus Nutans (Mal Bans) for Structural Application”
2. Supervisor of Master student (Nilkantha Regmi) for “Numerical optimization of opening in stone masonry homestays for seismic safety considering traditional architecture and comfort”
3. Co-Supervisor of Master student (Binita Silwal) for “Green treatment to improve resistance of Bambusa nutans and Bambusa balcooa against fungal degradation and UV exposure”
4. Supervisor of MoEST grant research on “Sustainable and Resilient Reconstruction of Traditional Stone Masonry Buildings in Mud Mortar.

At Asian Institute of Technology (AIT) (Co-supervise with Dr. Chaitanya Krishna)

1. Experimental study of soft-storey retrofitting of a typical reinforced concrete masonry infill frame based on numerical optimization, Master Student: Lin Htet (Completed)
2. Numerical analysis of seismic pounding of buildings in semi-urban areas due to out-of phase vibrations in adjacent buildings, Master Student: Prajwal Neupane
3. Numerical study of non-engineering defects on the modal properties of buildings using the 3D Applied Element Method, Master Student: Paweeta Wintachai
4. Data-driven numerical model updating of reinforced concrete structures using artificial neural networks (ANN), Master Student: Mr. Peerawut Watsaratiyanont
5. CUDA-based solver in AEM for large deformation of extra-large structures, Master Student: Mr. Phoowanun Tunnikorn

At Dhaka University (Co-supervise with Dr. Hossain Md. Shakhawat)

1. Numerical Analysis of Pounding Effects on Adjacent Buildings in Dhaka, Master Student: Malik Al Hasan Shuvo

Teaching Courses:

1. FBM-EL-577 Applied Element Modelling of Structures
2. SRI-EL-579 Finite Element Modelling of Structures
3. SRI-EL-583 Structural Evaluation and Retrofitting Methods for Existing Structures
4. SRI-EL-578 Advance Structural Dynamics and Vibration Control
5. FBM-CR- 501 Fundamentals of Forest Biomaterials Science (Wood Mechanics)

CURRENT PROJECTS AND FUNDING:

1. “Sustainable and Resilient Reconstruction of Traditional Stone Masonry Buildings in Mud Mortar” with Ministry of Education, Science and Technology (MoEST), February-June 2025: NRs. 670,000 **(Completed)**
2. “Characterization of Bambusa balcooa (Dhanu Bans) and Bambus nutans (Mal Bans) for Structural Application” with Base Bahay Foundation, 2024-2026: 1,060,350.00 Philippines Peso (PHP): Approx. NRs. 25 Lakhs
3. Partnership for “Research and Development in bamboo related research” with Green Bamboo Creation (GBC), 2025-2029 (5 years): NRs. 10 Lakhs.

Other Involvements:

- Member, Technical Committee for Fire Damage Assessment Guideline Development following the 2025 Gen-Z Protest Fire, Department of Urban Development and Building Construction (DUDBC), September, 2025
- Member, Research and Publication Subcommittee, Structural Engineers Association Nepal (SEANep), 2025

- Member, Technical Committee for Adoption of ISO Standards for Bamboo Design Guidelines, Nepal Bureau of Standards and Metrology (NBSM), 2025.

REFERENCE

Professor Kimiro Meguro,

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Dr. Muneyoshi Numada

Associate Professor, Department of Civil Engineering, The University of Tokyo, Japan, Email:

numa@iis.u-tokyo.ac.jp

Dr. Chaitanya Krishna Gadagamma,

Asst. Professor and Chair, Department of Structural Engineering, Asian Institute of Technology (AIT), Thailand

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